

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

What is the value of $\tan \frac{92\pi}{3}$?

Answer

- A. $-\sqrt{3}$
- B. $-\frac{\sqrt{3}}{3}$
- C. $\frac{\sqrt{3}}{3}$
- D. $\sqrt{3}$

Correct Answer: A

Rationale

Choice A is correct. A trigonometric ratio can be found using the unit circle, that is, a circle with radius **1** unit. If a central angle of a unit circle in the xy -plane centered at the origin has its starting side on the positive x -axis and its terminal side intersects the circle at a point (x, y) , then the value of the tangent of the central angle is equal to the y -coordinate divided by the x -coordinate. There are 2π radians in a circle. Dividing $\frac{92\pi}{3}$ by 2π yields $\frac{92}{6}$, which is equivalent to $15 + \frac{2}{3}$. It follows that on the unit circle centered at the origin in the xy -plane, the angle $\frac{92\pi}{3}$ is the result of **15** revolutions from its starting side on the positive x -axis followed by a rotation through $\frac{2\pi}{3}$ radians. Therefore, the angles $\frac{92\pi}{3}$ and $\frac{2\pi}{3}$ are coterminal angles and $\tan(\frac{92\pi}{3})$ is equal to $\tan(\frac{2\pi}{3})$. Since $\frac{2\pi}{3}$ is greater than $\frac{\pi}{2}$ and less than π , it follows that the terminal side of the angle is in quadrant II and forms an angle of $\frac{\pi}{3}$, or 60° , with the negative x -axis. Therefore, the terminal side of the angle intersects the unit circle at the point $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$. It follows that the value of $\tan(\frac{2\pi}{3})$ is $\frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}$, which is equivalent to $-\sqrt{3}$. Therefore, the value of $\tan(\frac{92\pi}{3})$ is $-\sqrt{3}$.

Choice B is incorrect. This is the value of $\frac{1}{\tan(\frac{92\pi}{3})}$, not $\tan(\frac{92\pi}{3})$.

Choice C is incorrect. This is the value of $\frac{1}{\tan(\frac{\pi}{3})}$, not $\tan(\frac{92\pi}{3})$.

Choice D is incorrect. This is the value of $\tan(\frac{\pi}{3})$, not $\tan(\frac{92\pi}{3})$.